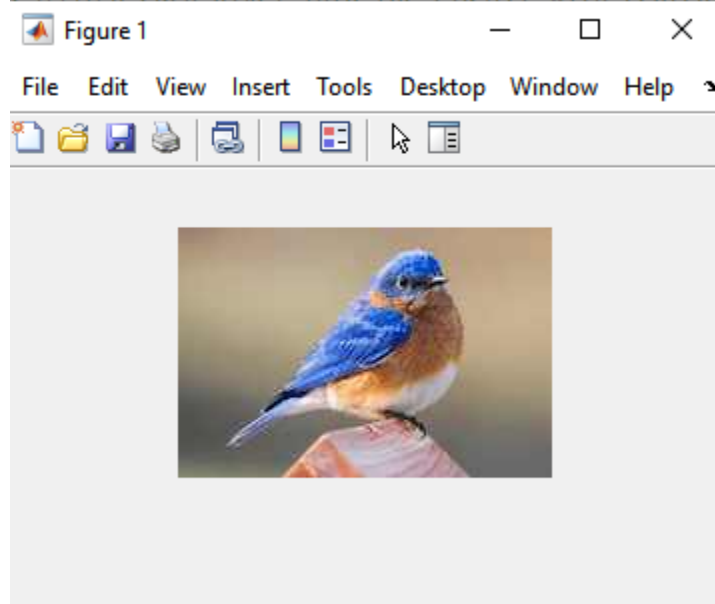
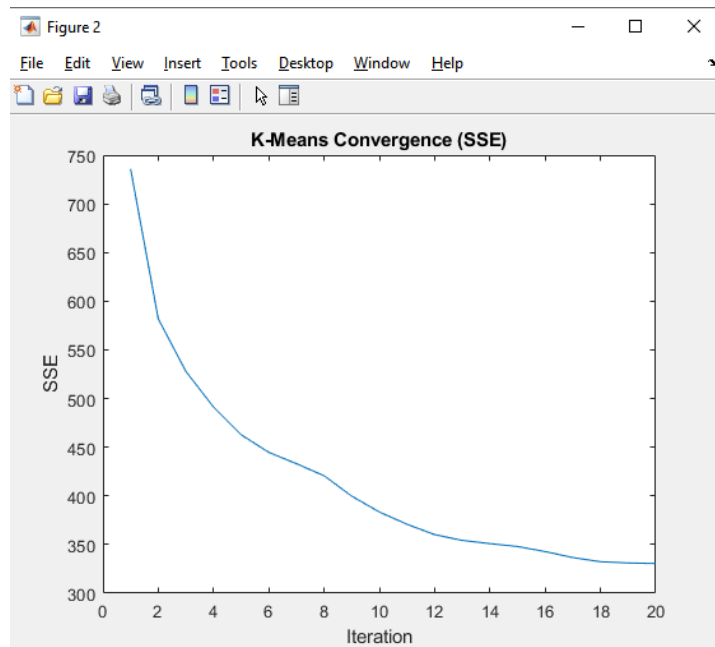


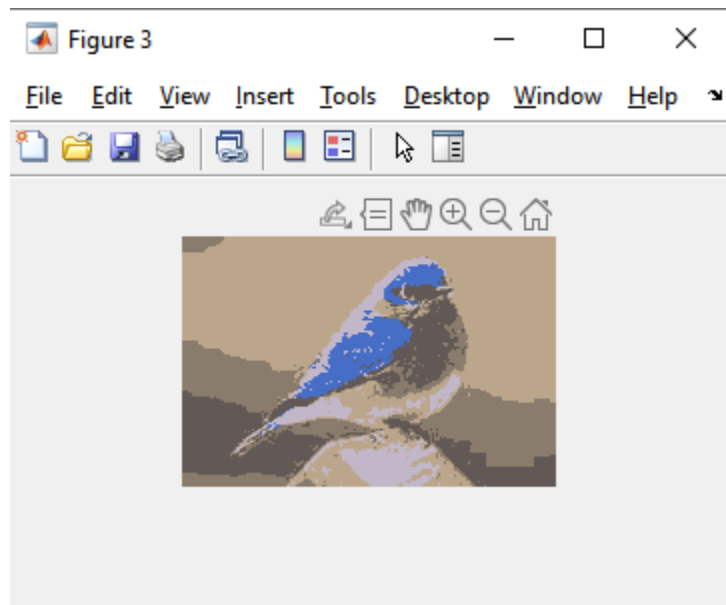
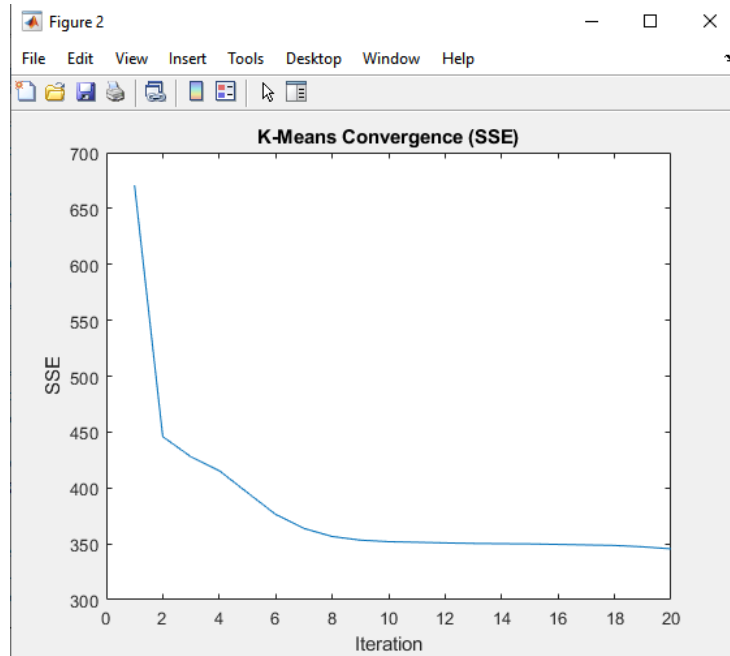
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Rhilo Sotto
11/23/2025

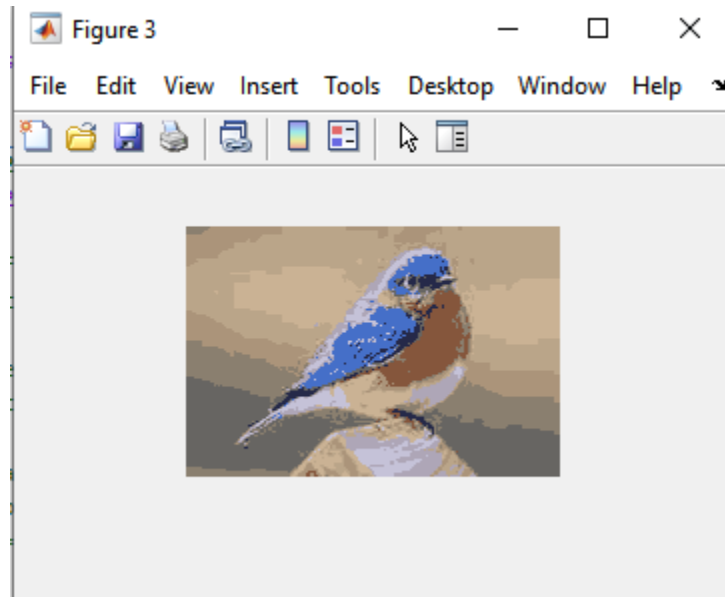
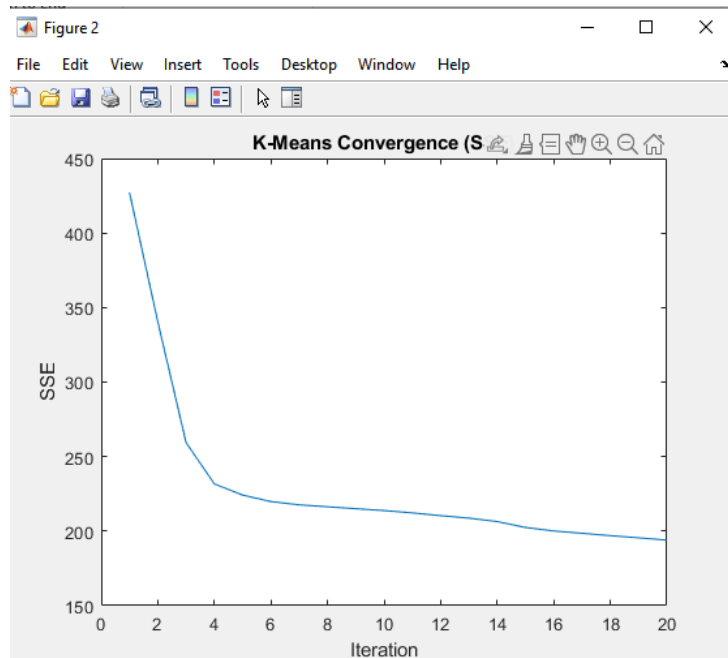
Graph of SSE



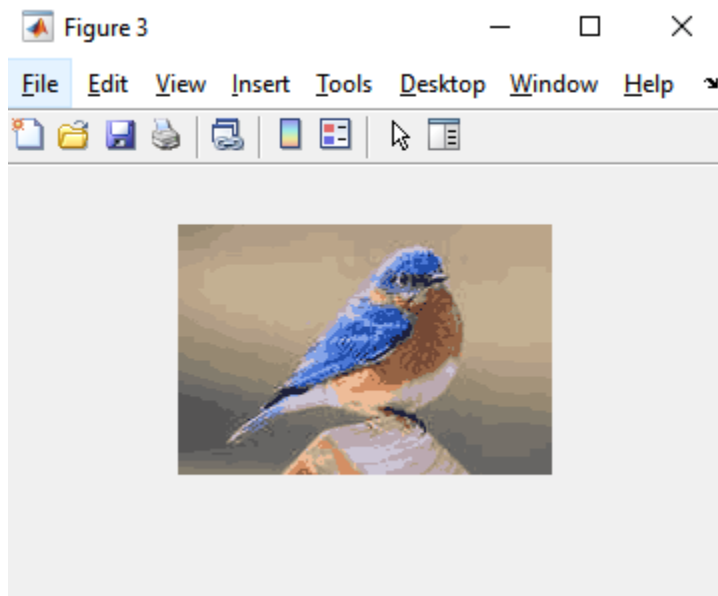
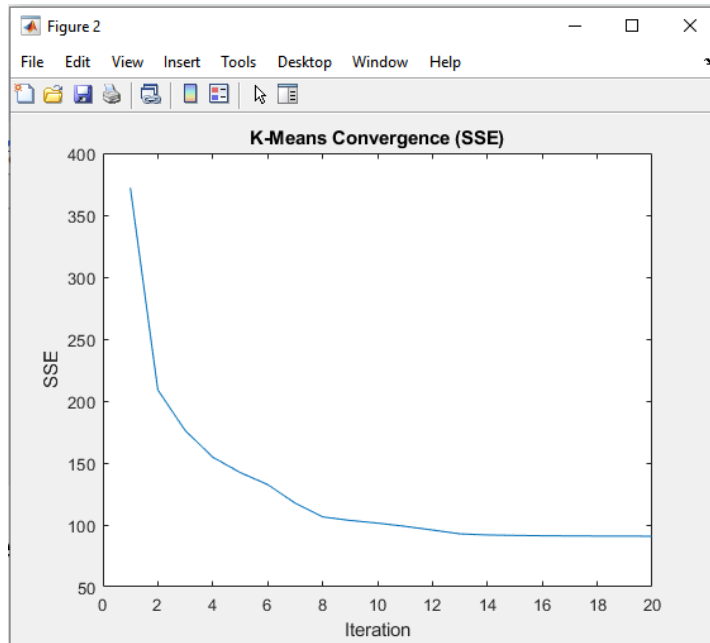
Compressed Image - K = 5



Compressed Image - $K = 10$



Compressed Image - K = 20



Questions:

1. With higher K values, the higher quality of the image, and also results in longer convergence times.
2. The K-means clustering algorithm produced visually better compressed images, where the colors are more varied. It was also easier to implement the clustering algorithm. PCA

however, preserves structure better as less data is lost from reconstructing the image compared to clustering.

3. Implementing went well for the K-means clustering, it was relatively simple to implement compared to PCA.